Adi Sharma

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Education

Sikkim Manipal Institute of Technology - SMU, Sikkim **2020 – 2024**
Bachelor of Technology - Computer Science and Engineering *CGPA: 8.53*

Coursework: Software Engineering, Computer Networks, Operating System, Theory of Computation, Object oriented programming, Database Management System, Data Structures, Design & Analysis of Algorithms, Distributed Systems, Artificial Intelligence, Machine Learning, Blockchain, Advance Web Technologies, AGILE, Parallel Programming, UI/UX Design, Cloud Computing, Discrete Mathematics

Technical Skills

Languages: Python, Go, JavaScript/TypeScript, C++, C, PowerShell, SQL
ML/AI Frameworks: TensorFlow, PyTorch, Scikit-learn, XGBoost, Sentence-BERT, BERTopic, spaCy, LangChain, LlamaIndex, MLflow, OpenCV
Backend Frameworks: FastAPI, Flask, Node.js, Express.js, GraphQL, ASP.NET Core 8.0
Frontend Frameworks: React.js (Hooks, ATOM Architecture, State Management)
Databases: PostgreSQL, TimescaleDB, MongoDB, Redis, MySQL, Greenplum, SQL Server
Cloud & Infrastructure: Azure (Data Factory, Synapse, ML, OpenAI, DevOps), Docker, Kubernetes (AKS, PKS), Terraform, PCF, Google Cloud
Message Queues & Search: RabbitMQ, Elasticsearch, SignalR
DevOps & CI/CD: GitLab CI/CD, Azure DevOps, Docker, Kubernetes, Git
Robotics & Simulation: ROS, Gazebo, ArduPilot, Arduino IDE
Analytics & Visualization: Power BI, Tableau, NumPy, Pandas
Development Tools: VS Code, Git, JIRA, Postman, DBeaver, Anaconda, Figma, Linux, Windows
Soft Skills: Teamwork, Leadership, Communication, Problem-Solving, Agile Methodology

Experience

Dell Technologies **Jan 2026 – Present**
Software Development Engineer 2 *Bengaluru, KA, India*

- Built real-time pub/sub system using SignalR and WebSockets to manage **100K+ devices globally** with **<3s command execution latency** and **99.7% success rate**, handling **720+ concurrent connections per server** across AMER/EMEA/APJ regions.
- Designed hub architecture with automatic reconnection, connection pooling, and group-based routing to maintain stable connections with 10s keepalive intervals and 30s timeouts, supporting up to 50MB message payloads for log transfers.
- Developed Native AOT .NET agent that reduced package size from 350MB to 30MB (**91% reduction**), memory usage from 200MB to 60MB, and installation time from 10 minutes to 30 seconds, saving **$67K annually** in bandwidth costs.
- Implemented targeted command dispatch with multi-status responses (SUCCESS/PARTIAL_SUCCESS/FAILURE) and 207 status codes for partial operations, enabling selective device management with full acknowledgment tracking.

Dell Technologies **Aug 2024 – Dec 2025**
Software Development Engineer 1 *Bengaluru, KA, India*

- Built ML-powered personalization engine for Dell Communique ($5M platform, **100K+ users**) using K-Means segmentation, XGBoost behavioral models (**78% accuracy**), and Sentence-BERT embeddings, improving open rates from 18% to 25% and click rates from 2.3% to 4.1% (**40% engagement lift**).
- Developed predictive analytics using Cox regression (**0.82 AUC-ROC**) for attrition analysis, propensity score matching for ROI measurement, and Isolation Forest for anomaly detection (**88% precision, <5min latency**), integrated with Azure Monitor for alerting.
- Built Go microservices for SharePoint integration that reduced API latency from 850ms to 180ms (**79% faster**) and increased throughput from 45 to 320 req/sec using goroutine pooling, HTTP/2 multiplexing, and optimized buffer handling.
- Implemented RabbitMQ email pipeline processing **10K emails/min** with **99.9% delivery rate** using SMTP connection pooling (20 connections), batch sending (100 emails/batch), and adaptive rate limiting.
- Designed Software Inventory Agent scanning engine for **100K+ devices** combining PowerShell registry queries and Python file analysis, reducing daily data collection from 2.4MB to 0.9MB per device (**62% reduction**).
- Built software normalization engine using fuzzy matching (Levenshtein distance) and TF-IDF similarity with **75% accuracy**, processing **16K unique titles across 8GB data**, improving throughput from 2,400 to 16,000 evidences/hour through parallel processing.
- Contributed to patent-pending Agentic AI system (**USPTO filing authorized**) combining XGBoost risk prediction (**87% detection, up from 61%**), K-Means persona classification (**91% accuracy**), and Isolation Forest anomaly detection (**84% catch rate**) for software governance.
- Built RL-based adaptive scheduler using Q-learning to optimize scan timing, achieving **95% idle-period execution** and reducing CPU impact from 28% to 6% (**79% reduction**) by learning from user activity patterns.

Dell Technologies | **January 2024 – May 2024**
Software Development Engineer Intern (Winter) | *Bengaluru, KA, India*

- **Developed semantic search engine** using Elasticsearch and Sentence-BERT embeddings (all-MiniLM-L6-v2), improving **user satisfaction from 58% to 84% (+26 points)** and reducing queries per search from **8 to 2.3** through fine-tuning on **12K training queries** with BM25 + vector hybrid retrieval.
- **Built hybrid recommendation system** combining matrix factorization (ALS, rank=50) and content-based filtering (TF-IDF + metadata), achieving **78% CTR improvement (23%→41%)** and **54% NDCG@5 gain** on training platform serving **100K+ users** with MLflow for A/B testing.
- **Transformed Azure DevOps CI/CD pipeline** from **58% to 94% success rate (+36 points)** with **73% build time reduction (45min→12min)** through parallel job execution (3 agents), Docker layer caching (BuildKit), and automated rollback mechanisms with quality gates.
- **Implemented GraphQL API layer** for Digital People Training Portal, eliminating over-fetching and under-fetching with precise field selection, optimizing data querying with relational data models and foreign key constraints for improved performance.
- **Designed ATOM architecture** (Atomic Design) for Security Governance Web App frontend from scratch, creating modular, reusable component library with dynamic API integration, enhancing scalability and maintainability across the application.

Dell Technologies | **May 2023 – July 2023**
Software Development Engineer Intern (Summer) | *Bengaluru, KA, India*

- **Engineered User360 dashboard** with FastAPI backend and React frontend for TMX Data Platform, reducing **API latency by 88% (3.2s→0.4s)** through query optimization (EXPLAIN ANALYZE), Redis caching (TTL=300s, **76% hit rate**), and connection pooling (min=5, max=20 connections).
- **Optimized data retrieval for 500K records by 96% (45s→1.8s)** using table partitioning (monthly), cursor-based pagination (limit=1000), parallel query execution, and materialized views for aggregations on Greenplum and PostgreSQL databases.
- **Scaled API throughput by 580% (50→340 req/sec)** through B-tree indexing on user_id and timestamp columns, achieving linear scaling to **500 req/sec** and enabling seamless data access across the organization.
- **Streamlined GitLab CI/CD deployment workflows**, achieving **40% reduction in deployment time** with automated testing, build processes, and zero-downtime releases, resulting in 6-month winter internship extension and full-time offer.

Birla Institute of Technology and Science, Pilani | **June 2022 – August 2022**
Summer Intern | *Pilani, RJ, India*

- **Developed AI-based multi-robot coordination system** under Prof. Dr. Hari Om Bansal for heterogeneous fleet (rovers: 60Wh, drones: 74Wh) sharing limited charging infrastructure, solving autonomous resource management challenges in constrained environments.
- **Designed LSTM-based power load forecasting model** achieving **90%+ accuracy (MAE <5W, MAPE <10%)**, enabling precise battery drain prediction across varying mission profiles for real-time scheduling and resource allocation decisions.
- **Implemented Hungarian algorithm** for optimal task assignment integrated with dynamic charging optimization, achieving **92% mission completion rate** and **72% charging port utilization** while maintaining **90% energy efficiency** across the fleet.
- **Validated simulation-based system in ROS and Gazebo** with real-time telemetry, demonstrating that intelligent resource management requires orchestrating multiple algorithms (forecasting, scheduling, path planning) as a unified system for optimal performance.

Honors & Awards

Dell Innovation Award (2025) – Recognized for pioneering real-time SignalR agent for Software Governance System managing 100K+ devices with <3s latency.

Dell Extraordinary Award (2025) – Awarded for exceptional delivery of Software Inventory Agent managing 100K+ devices globally, reducing unauthorized software by **56% (4,100→1,800 titles)** with minimal operational impact.

Dell Bravo Award (2025) – Honored for delivering high-quality ML-powered personalization features on tight timelines for Dell Communique platform, achieving **40% engagement lift**.

Google Developer Student Club (GDSC) Lead (2022-2023) – Selected as one of **560 GDSC Leads across PAN India**, leading core team in organizing technical workshops and community events for 500+ students. Invited to GDSC Conference for leadership excellence.

Dell Hack-2-Hire Hackathon Winner (2022) – Developed computer vision model for object detection and natural language generation to convert images into human-readable descriptions, creating assistive technology for visually impaired individuals.

2nd Position, IETE Innovation Challenge (2019) – Awarded second position by Institution of Electronics and Telecommunication Engineers for developing autonomous FPV rover for ranging and surveillance, demonstrating expertise in embedded systems and autonomous navigation.

Leadership & Mentoring

Google Developer Student Club (GDSC) Lead (2022-2023) – Selected as one of **560 GDSC Leads across PAN India** for Sikkim's sole Google DSC. Led core team organizing coding workshops, webinars, study jams, and tech talks for **500+ students**. Invited to GDSC Conference for leadership excellence.

CS Student Council – General Secretary (2023) – Conducted technical workshops for **100+ students** on Computer Networks, Applied ML, Data Structures, and Algorithms, bridging theory and practical applications through hands-on sessions.

AI Club – ML Mentor (2021-2022) – Mentored **50+ students** in machine learning fundamentals (supervised/deep learning, model evaluation) through hands-on workshops using TensorFlow, Scikit-Learn, and PyTorch.

Service & Community

Google Developer Group (GDG) Bangalore – Core Member & Event Organizer (2024-Present) – Organized and coordinated end-to-end developer community meetups as EMCEE, facilitating knowledge sharing and networking for Bangalore's developer ecosystem with **200+ attendees per event**.

Technical Leadership Roles: Head of IT – TEDxSMIT | Head of IT – IAESTE SMU | Lead Tech Team – Smart India Hackathon (Government of India Initiative) | Core Member – NSS SMIT

Community Service & Social Impact: National Service Scheme (NSS) volunteer for 2023 Sikkim Flash Flood relief efforts, cleanliness drives, and AIDS awareness campaigns | CS Council outreach program teaching **50+ underprivileged school students** (2022)

Projects

DDoS Detection in SDN using Machine Learning | *Python, Scikit-learn, POX Controller, Mininet, OpenFlow*

- Developed an **intelligent intrusion detection system** leveraging machine learning to identify and mitigate DDoS attacks in Software-Defined Networks. Designed **real-time feature extraction** pipeline from OpenFlow statistics, extracting **23 discriminative features** with sliding window aggregation achieving **89ms detection latency**.
- Addressed **class imbalance** using SMOTE and cost-sensitive learning, systematically evaluated 5 ML algorithms across multiple attack types. Achieved **98.88% accuracy, 98.9% precision, and 98.85% recall** with Random Forest (n_estimators=150) after hyperparameter optimization across **180 configurations** using GridSearchCV.
- Integrated ML inference engine with **POX SDN controller** using asynchronous event-driven architecture and batch inference mechanism. Improved throughput from **15 to 560 flows/sec** and reduced latency by **96% (2-3s→89ms)** with zero packet drops during detection.
- Validated system on **Mininet topology** (10 switches, 50 hosts) achieving **98.95% detection rate** at 1000 pps attack intensity with **110ms mitigation time** and minimal legitimate traffic impact (**0.02% packet loss**). Implemented automated mitigation through dynamic flow rule installation.

Image Captioning System for Assistive Technology | *TensorFlow, Keras, Flask, ResNet50, VGG16, LSTM, Python*

- Designed an **end-to-end image captioning pipeline** combining pre-trained CNN encoders (ResNet50, VGG16, VGG19) with LSTM and Bi-LSTM decoders to generate natural-language descriptions for assistive use by visually impaired users.
- Conducted **systematic model comparison** across **four CNN-RNN architectures**, selecting the best configuration based on BLEU-based evaluation to balance caption fluency and relevance for optimal user experience.
- Achieved **BLEU-1 score of 0.61 and BLEU-2 score of 0.42** with the ResNet50 + Bi-LSTM model, demonstrating competitive performance relative to baseline captioning approaches on similar-scale datasets.
- Deployed the trained model as a **Flask web application** integrated with **text-to-speech** (pyttsx3), enabling users to upload images and receive spoken captions in real time for accessibility support.